BUILDING AI INFRASTRUCTURE · PART 5 OF 7

Part 5: Google Cloud — GKE and GPUs

Chapter summary for beginners

In one sentence: the same platform on Google Cloud: a private GKE cluster, a GPU pool where Google installs the driver for you, and reliability targets written as code.

Video length: about 9 minutes · **Code:** nawex-platform, this repo's `terraform/modules/gke` and `observability/slo` · **Cost:** often covered by trial credit

The 8 blocks on Google Cloud

#	Block	On Google Cloud
1	Compute	Compute Engine: general pool + GPU pool (**NVIDIA T4**)
2	Data and storage	Cloud Storage; **Artifact Registry** for images
3	Networking	VPC-native network, private nodes, **Cloud NAT**
4	Orchestration	**GKE**, in Standard or Autopilot mode
5	MLOps	MLflow + Kubeflow (or Vertex AI)
6	Serving	KServe + vLLM on the GPU pool
7	Observability	Cloud Monitoring, Managed Prometheus, **SLOs**
8	Guardrails	IAM, Workload Identity, Secret Manager, budgets

VPC-native means pods get real addresses from the network. **Private nodes** have no public IPs, and **Cloud NAT** lets them reach the internet on the way out only.

Three things before you build

1. **Budget:** Billing → Budgets & alerts. Set one even if you have trial credit.
2. **Two GPU quotas.** Google needs both *NVIDIA T4 GPUs* (in your region) **and** *GPUs (all regions)*. Request at least 1 of each; people often miss the global one.
3. **Log in twice:** `gcloud auth login` (for you) and `gcloud auth application-default login` (for Terraform). Then run `gcloud config set project <id>`.

Build it

```
cd nawex-platform
cp deploy/terraform/live/gcp/dev/terraform.tfvars.example
deploy/terraform/live/gcp/dev/terraform.tfvars
export GCP_PROJECT=my-project-id
GPU_ENABLED=1 scripts/deploy.sh gcp dev plan
GPU_ENABLED=1 scripts/deploy.sh gcp dev apply
kubectl get nodes -L cloud.google.com/gke-nodepool -L nvidia.com/gpu.present
```

With this repo: `terraform -chdir=terraform/environments/staging plan -var gpu_enabled=true`.

GPUs on GKE

The GPU pool sets `gpu_driver_version = DEFAULT`, so **GKE installs the NVIDIA driver at boot**. You still install the GPU Operator (with `driver.enabled=false`) for metrics and labels. Same vLLM manifest, same 0 → 1 → 0 behaviour.

Standard vs. Autopilot: in Autopilot, Google manages every node and bills per pod. It's simpler, but you get less control over GPUs. Standard is better for learning the full picture.

The big idea: SLOs as code

An **SLO** is a promise about reliability. Instead of alerting on every error, you alert on how fast you're spending the error budget.

Number	Meaning
99.5%	of requests succeed over 28 days: the availability SLO. That leaves a 0.5% **error budget**.
14.4×	burn rate over 1 hour: you'd spend about 2% of the month's budget in that hour. **Page someone now.**
6×	burn rate over 6 hours: a slow leak. **Open a ticket.**

This is the multi-window, multi-burn-rate method from Google's SRE Workbook. In this repo the SLOs live in `observability/slo/` (Sloth format). The video's Terraform version is in express-reliability-platform-course v10.

Cost and clean-up

GKE has a cluster management fee, but one zonal or Autopilot cluster per billing account falls under a free tier. You pay for nodes (GPUs most of all), Cloud NAT and the load balancer.

Tear down: `scripts/deploy.sh gcp dev destroy`. Then check for leftover **disks** and **forwarding rules**.

Words to know

- **Cloud NAT:** lets private machines reach the internet without being reachable from it.
- **Autopilot:** a GKE mode where Google runs the nodes for you.
- **Error budget:** the failure an SLO allows (100% minus the target).
- **Burn rate:** how fast you're spending the budget, compared with "just on target".
- **Page vs. ticket:** wake someone now, vs. fix it during working hours.

Check yourself

1. Which two GPU quotas does Google need?
2. Who installs the GPU driver on GKE?
3. With a 99.5% SLO, what's the error budget?
4. A 14.4× burn over 1 hour: page or ticket?

Answers: 1) NVIDIA T4 GPUs in your region, and GPUs (all regions). 2) GKE itself, at boot (`gpu_driver_version = DEFAULT`). 3) 0.5%. 4) Page.

Next: Part 6, the same platform on your own hardware, in a data center.

Emmanuel Naweji · emmanuelnaweji.com · transformed2succeed.com
Code: github.com/Here2ServeU/building-ai-infra · Series guide: docs/series/README.md